

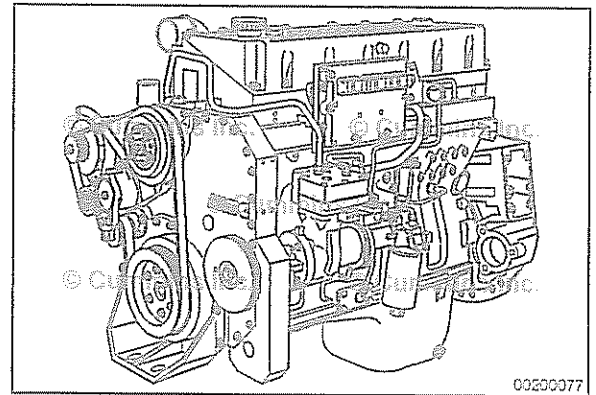
Engine Identification

Engine Dataplate

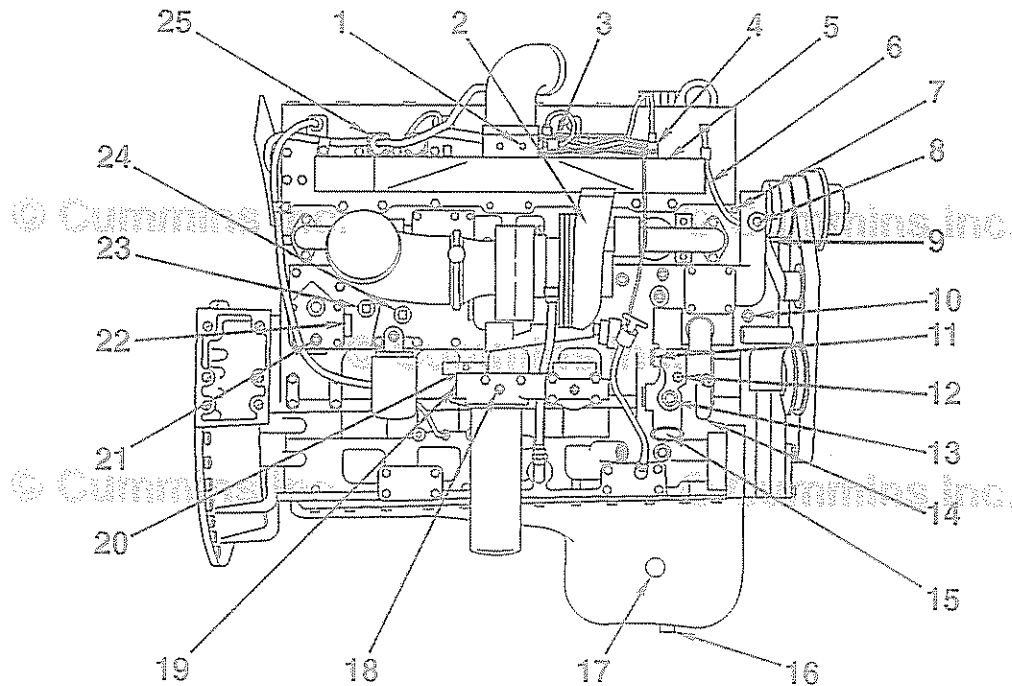
Marine Applications

The engine dataplate shows important facts about your engine. The engine serial number and CPL provide data for ordering parts and service. The engine dataplate must not be changed unless approved by Cummins Inc.

The dataplate is located on the fuel pump side of the engine, on the rocker housing. Have the following engine data available when communicating with a Cummins Authorized Repair Location. The following information on the dataplate is mandatory when sourcing service parts:



Industrial and Generator Drive

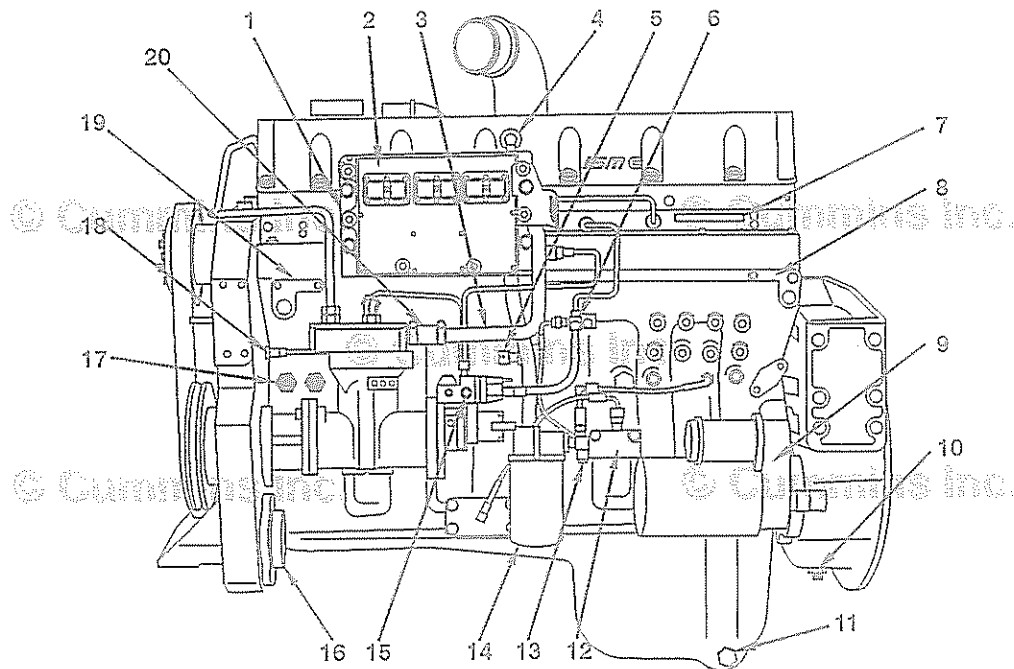


Industrial and Generator Drive - Exhaust Side

- | | |
|--|--------------------------------|
| 1. Ether hookup | 14. Water pump pressure |
| 2. Before charge air cooler | 15. Coolant inlet |
| 3. Boost temperature sensor | 16. Oil drain |
| 4. Intake manifold pressure sensor | 17. Oil pan pump heater |
| 5. After charge air cooler (intake manifold) | 18. Filter outlet pressure |
| 6. Shutter stat | 19. Engine brake oil supply |
| 7. Engine vent coolant | 20. Filter inlet pressure |
| 8. Shutter stat | 21. Block coolant pressure |
| 9. Fan sensor | 22. Coolant heater |
| 10. Coolant temperature sensor | 23. Heater supply |
| 11. Heater return | 24. Coolant temperature pickup |
| 12. Coolant inlet pressure | 25. Wastegate controller. |
| 13. Makeup line from radiator | |

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Industrial and Generator Drive

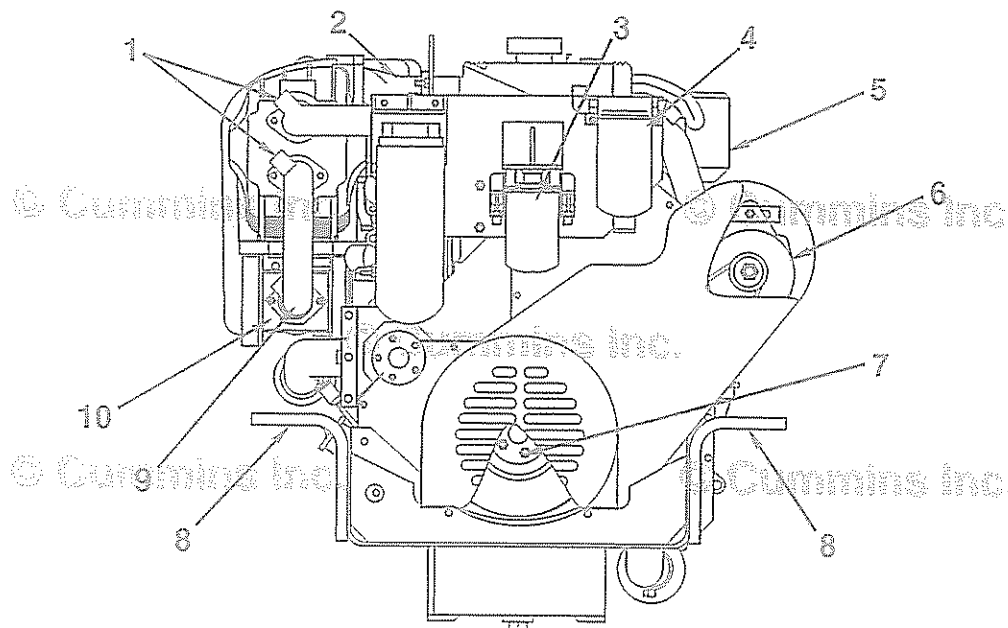


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Industrial and Generator Drive - Fuel Pump Side

- | | |
|-------------------------------------|--|
| 1. ECM cooling plate | 11. Side oil drain |
| 2. ECM | 12. Centinel™ (optional) |
| 3. Compressor air inlet | 13. Fuel inlet to pump |
| 4. Blowby measurement | 14. Fuel filter |
| 5. Ambient air pressure sensor | 15. Rail pressure |
| 6. Fuel return to tank | 16. Power steering pump mounting location |
| 7. Engine data tag | 17. Oil pressure and temperature sensor |
| 8. Engine serial number | 18. Engine position sensor |
| 9. Starter | 19. Freon compressor mounting location. |
| 10. Flywheel ring gear speed sensor | 20. Compressor air discharge (not visible) |

Marine Applications

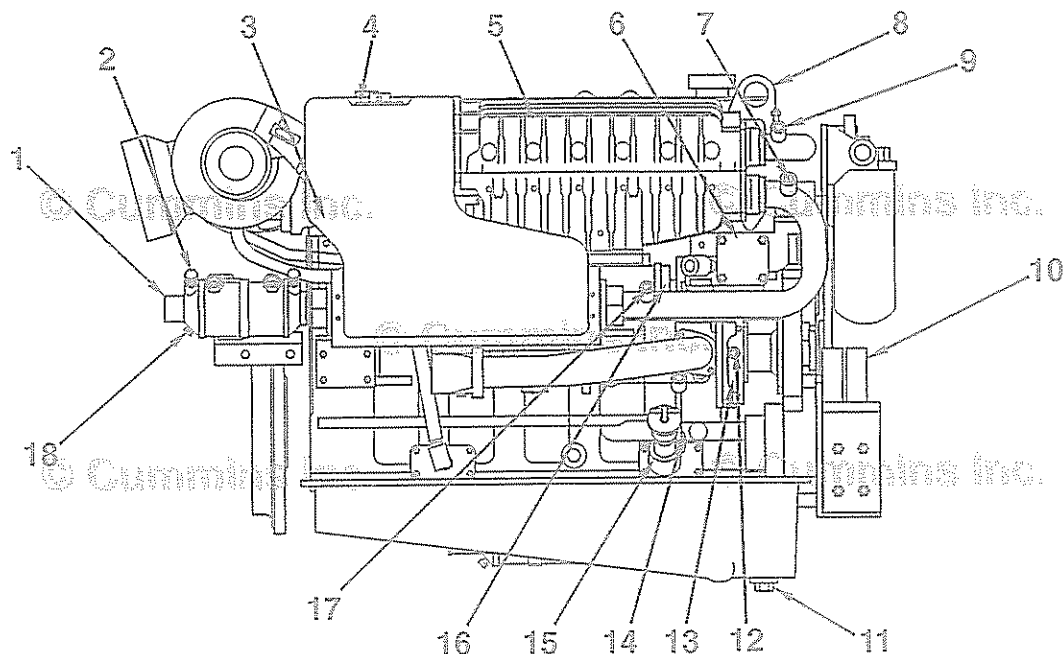


Marine - Front View

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- | | |
|-----------------------|---------------------------------------|
| 1. Zinc plugs | 6. Alternator |
| 2. Air crossover | 7. Vibration damper |
| 3. Coolant filter | 8. Front engine mounts |
| 4. Fuel Filter | 9. Aftercooler-to-heat-exchanger pipe |
| 5. Crankcase breather | 10. Heat exchanger. |

Marine Applications

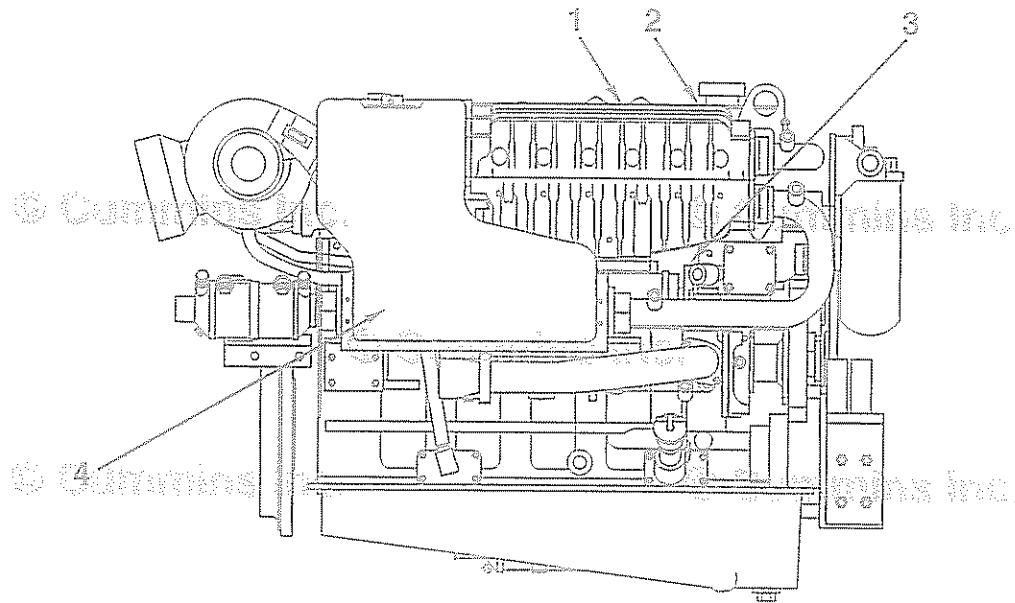


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Marine - Starboard View

- | | |
|-----------------------------|---------------------------------|
| 1. Gear oil cooler | 10. Vibration damper |
| 2. Zinc anode (gear cooler) | 11. Oil pan drain plug |
| 3. Heat exchanger | 12. Coolant drain |
| 4. Expansion tank | 13. Water pump |
| 5. Aftercooler | 14. Oil dipstick |
| 6. Thermostat housing | 15. Oil fill |
| 7. Zinc anode | 16. Engine oil cooler |
| 8. Engine lifting bracket | 17. Zinc anode |
| 9. Zinc anode | 18. Gear oil cooler drain plug. |

Marine Applications

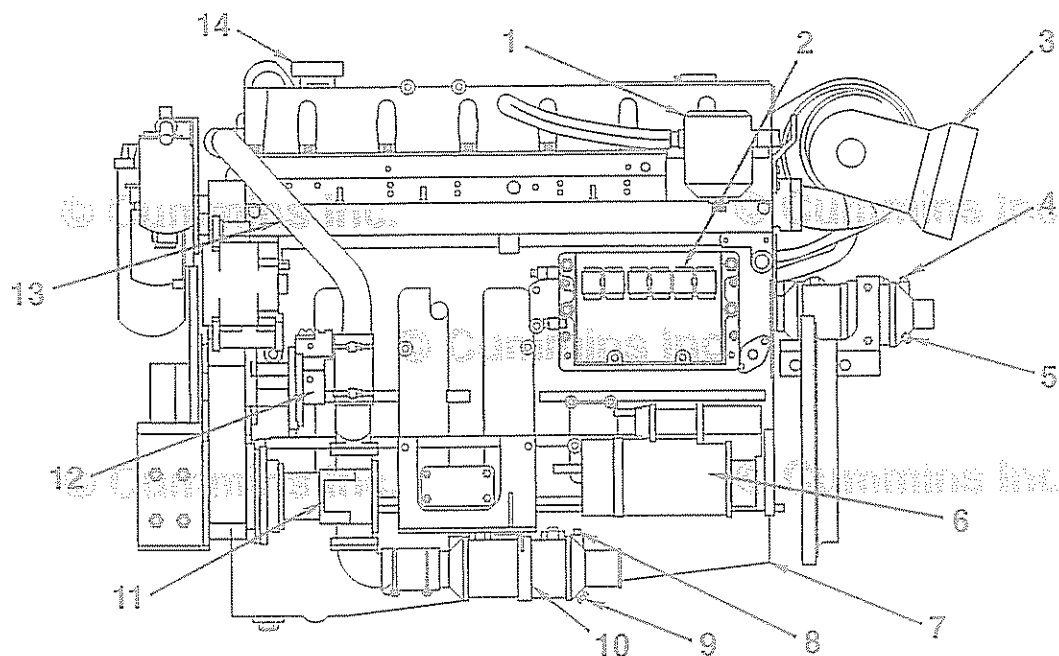


QSM11 Starboard View

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1. Intake manifold temperature sensor
2. Intake manifold pressure sensor
3. Coolant temperature sensor
4. Coolant level sensor

Marine Applications

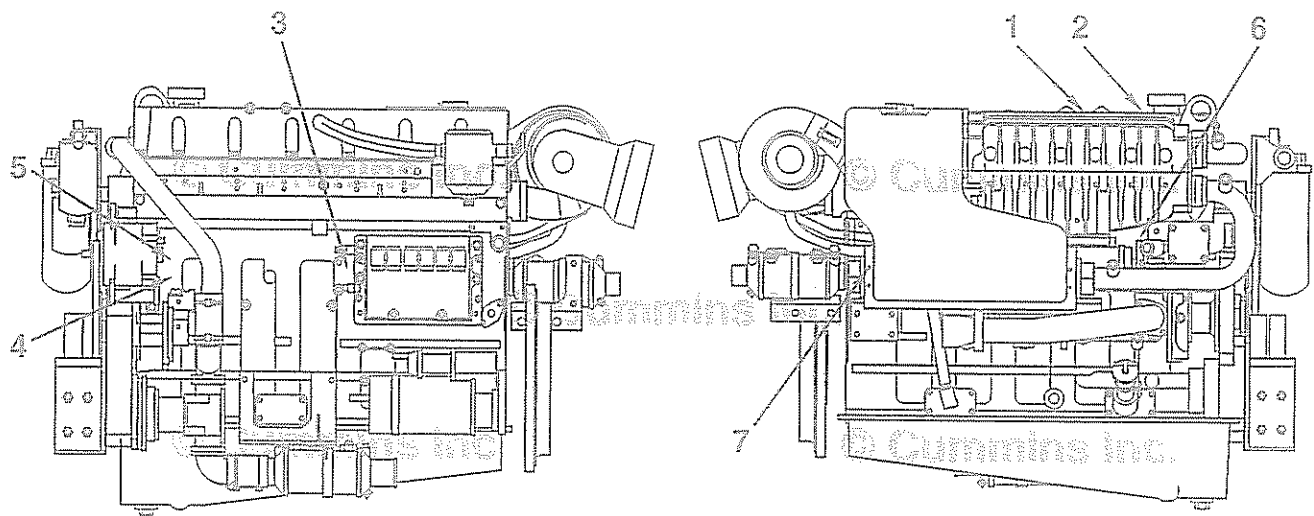


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Marine - Port View

- | | |
|---------------------------|--|
| 1. Crankcase breather | 8. Zinc anode (fuel cooler) |
| 2. ECM | 9. Fuel cooler drain plug |
| 3. Exhaust elbow | 10. Fuel cooler |
| 4. Zinc anode | 11. Sea water pump |
| 5. Gear cooler drain plug | 12. Fuel pump |
| 6. Starter | 13. Sea-water-pump-to-aftercooler pipe |
| 7. Oil pan | 14. Oil fill. |

Marine Applications

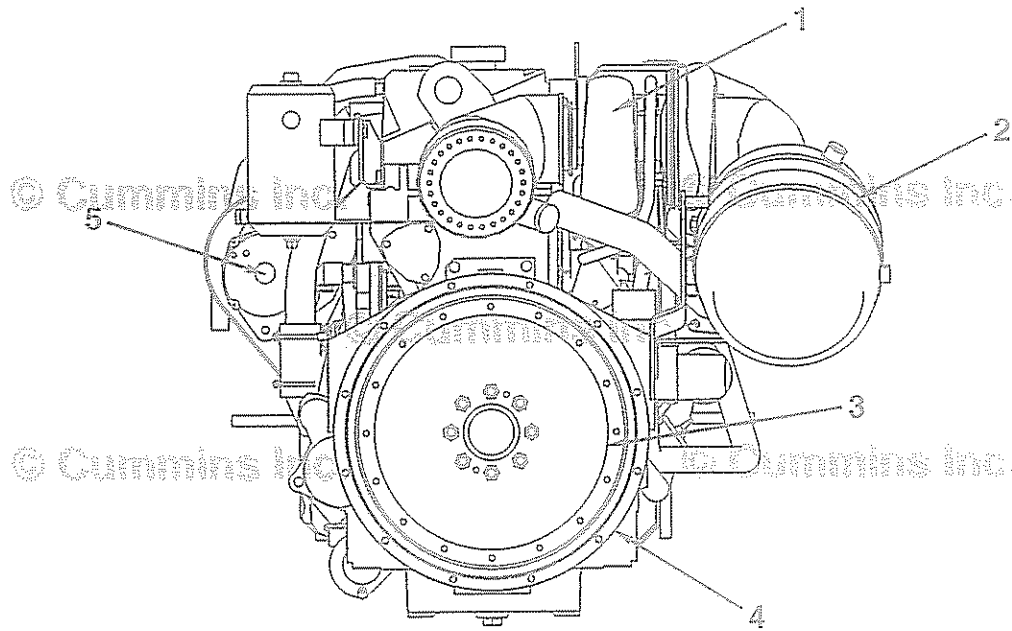


QSM 11 Port View

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1. Ambient air pressure sensor
2. Oil pressure/temperature sensor (behind fuel pump)
3. Engine position sensor (behind fuel pump)

Marine Applications



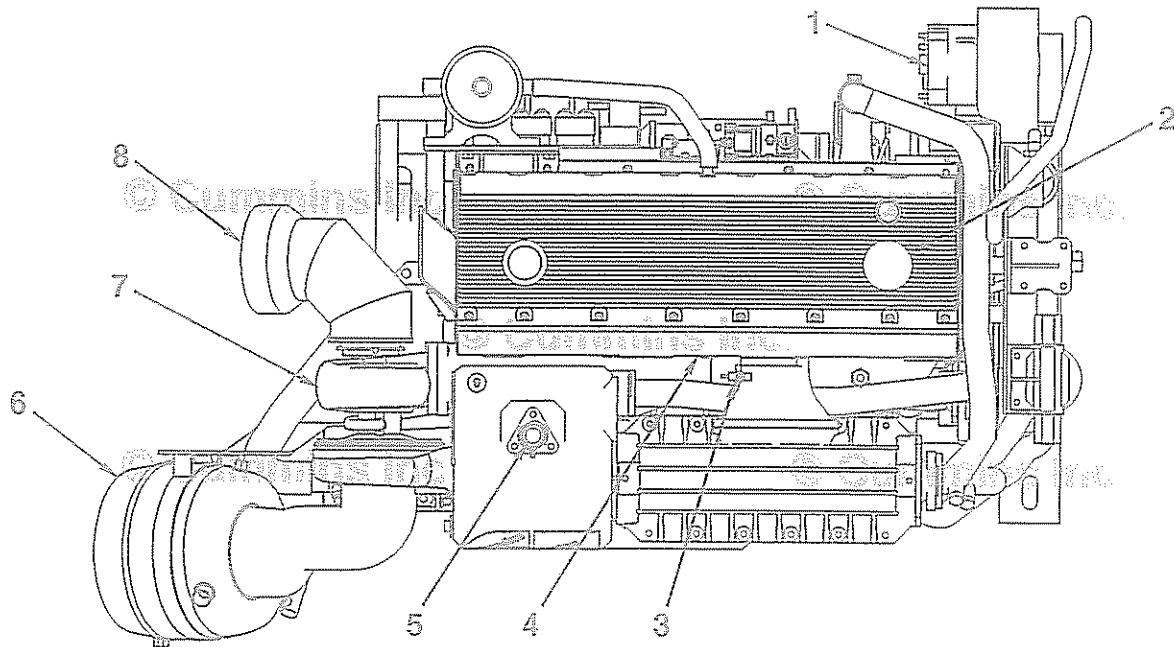
Marine - Rear View

- 1. Turbocharger
- 2. Air cleaner
- 3. Flywheel

- 4. Flywheel housing
- 5. Alternator.

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Marine Applications

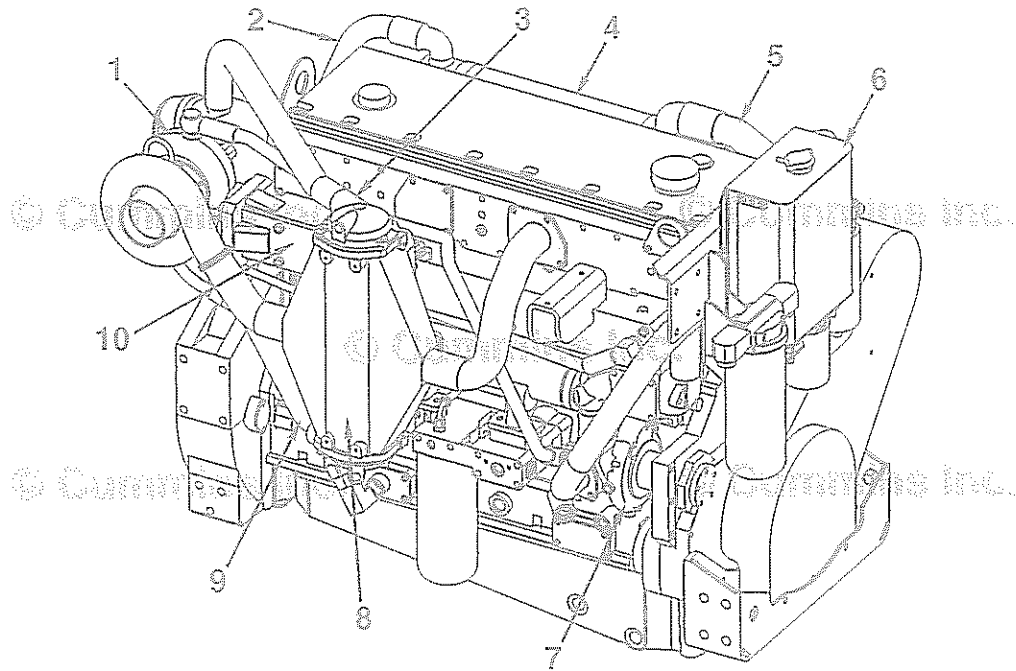


17200016

Marine - Top View

- | | |
|---------------------------------------|-------------------|
| 1. Alternator | 5. Coolant fill |
| 2. Oil fill (alternate) | 6. Air cleaner |
| 3. Intake manifold temperature sensor | 7. Turbocharger |
| 4. Intake manifold pressure sensor | 8. Exhaust elbow. |

Marine Applications

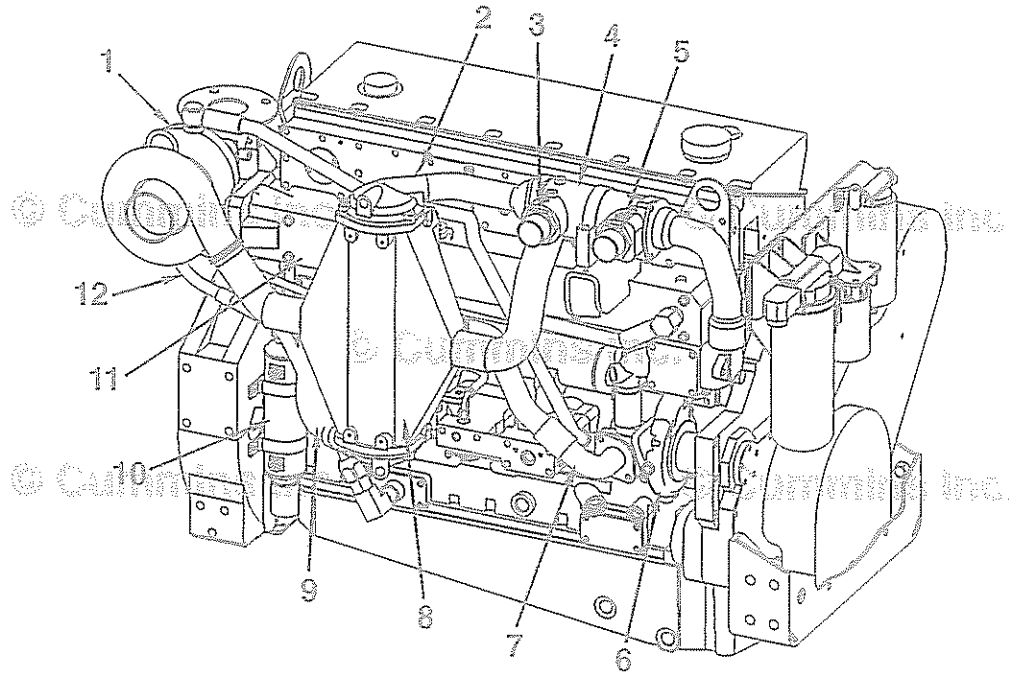


Heat Exchanger

1. Turbocharger with water cooled turbine housing
2. Coolant flow from heat exchanger
3. Aftercooler sea water outlet
4. Heat exchanger
5. Coolant flow to heat exchanger
6. Expansion tank
7. Water pump
8. Aftercooler
9. Aftercooler sea water inlet
10. Water cooled exhaust manifold

08206244

Marine Applications



Keel Cooled

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1. Turbocharger with water cooled turbine housing
2. Coolant inlet (aftercooler)
3. Coolant in from Keel Cooler
4. LTA thermostat housing
5. Coolant out to Keel Cooler
6. Water pump
7. Coolant to water pump
8. Aftercooler
9. Coolant outlet (aftercooler) to water pump
10. Gear oil cooler (optional)
11. Water cooled exhaust manifold
12. Coolant out from turbocharger

General Engine

Specifications

Industrial and Generator Drive

Type.....Four-cycle, In-line six-cylinder, four-valves per cylinder
Horsepower.....Refer to engine dataplate
Engine Speed at Maximum Output.....Refer to engine dataplate
Bore and Stroke.....125 mm [4.921 in] x 147 mm [5.787 in]
Displacement.....10.8 liters [659 C.I.D.]
Firing Order.....1-5-3-6-2-4
Engine Weight (with standard accessories):.....940 kg [2072 lb]

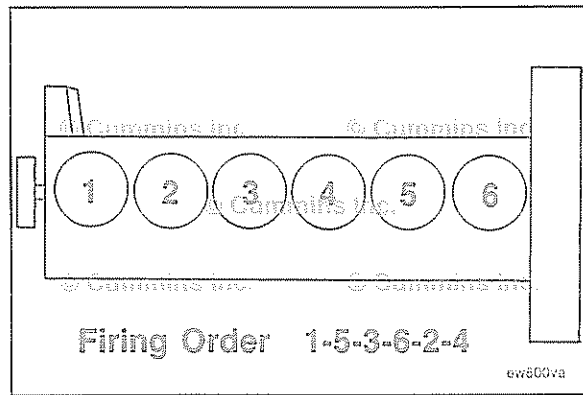
Industrial Applications

Generator Drive Applications

Crankshaft Rotation (viewed from the front of the engine)..... Clockwise

Marine Applications

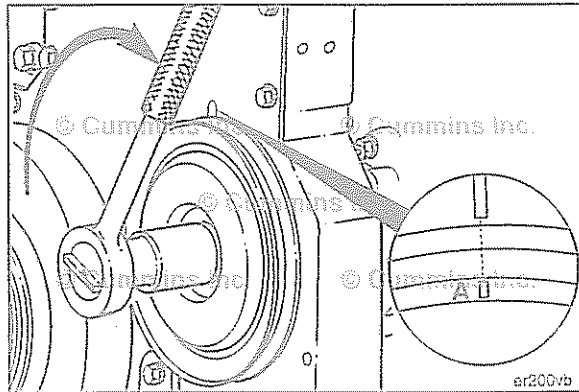
Type.....Four-cycle, In-line six-cylinder, four-valves per cylinder
Horsepower.....Refer to engine dataplate
Engine Speed at Maximum Output.....Refer to engine dataplate
Bore and Stroke.....125 mm [4.921 in] x 147 mm [5.787 in]
Displacement.....10.8 liters [659 C.I.D.]
Firing Order.....1-5-3-6-2-4
Engine Weight (with standard accessories):.....1124 kg [2478 lb]
Dry Weight.....1124 kg [2478 lb]
Crankshaft Rotation (viewed from the front of the engine)..... Clockwise



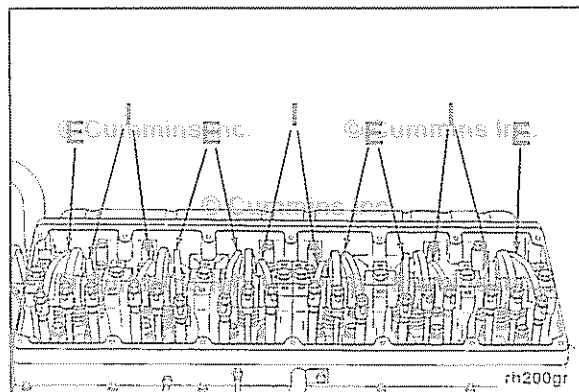
The crankshaft rotation is clockwise when viewed from the front of the engine.

The cylinders are numbered from the front end of the engine.

The firing order is 1-5-3-6-2-4.



Rotate the accessory drive clockwise until the A valve set mark on the accessory drive pulley is aligned with the pointer on the gear cover.



Each cylinder has three rocker levers:

- The long rocker lever (E) is the exhaust lever.
- The center rocker lever is the injector lever
- The short rocker lever (I) is the intake lever.

Exhaust System

Specifications

Marine Applications

Exhaust Pipe Size (normally acceptable inside diameter).....102 mm [4 in]

Maximum Exhaust Pipe Restriction

H₂O.....1016 mm H₂O [40 in H₂O]

Hg.....75 mm Hg [3 in Hg]

Industrial and Generator Drive

Exhaust Pipe Size (normally acceptable inside diameter).....102 mm [4 in]

Maximum Exhaust Pipe Restriction

H₂O.....1016 mm H₂O [40 in H₂O]

Hg.....75 mm Hg [3 in Hg]

Compressed Air System

Specifications

Cummins 18.7 CFM Model

Cylinders.....	1
Compressor Swept Volume at 1250.....	6.6 l/sec. [140 cfm]
Piston Displacement.....	318 cc [19.405 C.I.D.]
Bore.....	85 mm [3.346 in]
Stroke.....	56 mm [2.205 in]
Speed.....	Engine speed
Cooling.....	Engine coolant
Lubrication.....	Engine lubricating oil
Plumbing Line Sizes:	
Coolant Inlet and Outlet.....	3/4 x 16 STOR
Air Inlet.....	1-in nominal hose
Air Outlet.....	M27 x 2 STOR
Unloader Port.....	M10 x 1 STOR
Governor Mounting Direct.....	M8 x .25
Height, Overall (approximate).....	217.4 mm [8.559 in]
Width, Overall (approximate).....	142 mm [5.591 in]
Length, Overall (approximate).....	216 mm [8.504 in]
Weight (approximate).....	15 kg [33.0 lb]

Cummins 37.4 CFM Model

Cylinders.....	1
Compressor Swept Volume at 1250.....	15.5 l/sec. [35.5 cfm]
Piston Displacement.....	636 cc [39 C.I.D.]
Bore.....	85 mm [3.35 in]
Stroke.....	56 mm [2.2 in]
Speed.....	Engine speed
Cooling.....	Engine coolant
Lubrication.....	Engine lubricating oil
Plumbing Line Sizes:	
Coolant Inlet and Outlet.....	3/4 x 16 STOR
Air Inlet.....	1-in nominal hose
Air Outlet.....	M27 x 2 STOR
Unloader Port.....	M10 x 1 STOR
Governor Mounting Direct.....	M8 x .25
Height, Overall (approximate).....	306 mm [12 in]
Width, Overall (approximate).....	142 mm [5.6 in]
Length, Overall (approximate).....	286.7 mm [11.29 in]
Weight (approximate).....	17.24 kg [38 lb]